

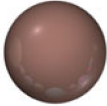
Appendix II:

Useful Data

A. Atomic Colors

Atomic number:	1	4	5	6	7	8	9
							
Atomic symbol:	H	Be	B	C	N	O	F

Atomic number:	11	12	14	15	16	17	19
							
Atomic symbol:	Na	Mg	Si	P	S	Cl	K

Atomic number:	20	29	30	35	53	54
						
Atomic symbol:	Ca	Cu	Zn	Br	I	Xe

B. Standard Thermodynamic Quantities for Selected Substances at 25 °C

Substance	ΔH_f° (kJ/mol)	ΔG_f° (kJ/mol)	S° (J/mol · K)
Aluminum			
Al(s)	0	0	28.32
Al(g)	330.0	289.4	164.6
Al ³⁺ (aq)	−538.4	−483	−325
AlCl ₃ (s)	−704.2	−628.8	109.3
Al ₂ O ₃ (s)	−1675.7	−1582.3	50.9
Barium			
Ba(s)	0	0	62.5
Ba(g)	180.0	146.0	170.2
Ba ²⁺ (aq)	−537.6	−560.8	9.6
BaCO ₃ (s)	−1213.0	−1134.4	112.1
BaCl ₂ (s)	−855.0	−806.7	123.7
BaO(s)	−548.0	−520.3	72.1
Ba(OH) ₂ (s)	−944.7		
BaSO ₄ (s)	−1473.2	−1362.2	132.2

Substance	ΔH_f° (kJ/mol)	ΔG_f° (kJ/mol)	S° (J/mol · K)
Beryllium			
Be(s)	0	0	9.5
BeO(s)	−609.4	−580.1	13.8
Be(OH) ₂ (s)	−902.5	−815.0	45.5
Bismuth			
Bi(s)	0	0	56.7
BiCl ₃ (s)	−379.1	−315.0	177.0
Bi ₂ O ₃ (s)	−573.9	−493.7	151.5
Bi ₂ S ₃ (s)	−143.1	−140.6	200.4
Boron			
B(s)	0	0	5.9
B(g)	565.0	521.0	153.4
BCl ₃ (g)	−403.8	−388.7	290.1
BF ₃ (g)	−1136.0	−1119.4	254.4
B ₂ H ₆ (g)	36.4	87.6	232.1

(continued on the next page)

Substance	ΔH_f° (kJ/mol)	ΔG_f° (kJ/mol)	S° (J/mol · K)
B ₂ O ₃ (s)	−1273.5	−1194.3	54.0
H ₃ BO ₃ (s)	−1094.3	−968.9	90.0
Bromine			
Br(g)	111.9	82.4	175.0
Br ₂ (l)	0	0	152.2
Br ₂ (g)	30.9	3.1	245.5
Br [−] (aq)	−121.4	−102.8	80.71
HBr(g)	−36.3	−53.4	198.7
Cadmium			
Cd(s)	0	0	51.8
Cd(g)	111.8	77.3	167.7
Cd ²⁺ (aq)	−75.9	−77.6	−73.2
CdCl ₂ (s)	−391.5	−343.9	115.3
CdO(s)	−258.4	−228.7	54.8
CdS(s)	−161.9	−156.5	64.9
CdSO ₄ (s)	−933.3	−822.7	123.0
Calcium			
Ca(s)	0	0	41.6
Ca(g)	177.8	144.0	154.9
Ca ²⁺ (aq)	−542.8	−553.6	−53.1
CaC ₂ (s)	−59.8	−64.9	70.0
CaCO ₃ (s)	−1207.6	−1129.1	91.7
CaCl ₂ (s)	−795.4	−748.8	108.4
CaF ₂ (s)	−1228.0	−1175.6	68.5
CaH ₂ (s)	−181.5	−142.5	41.4
Ca(NO ₃) ₂ (s)	−938.2	−742.8	193.2
CaO(s)	−634.9	−603.3	38.1
Ca(OH) ₂ (s)	−985.2	−897.5	83.4
CaSO ₄ (s)	−1434.5	−1322.0	106.5
Ca ₃ (PO ₄) ₂ (s)	−4120.8	−3884.7	236.0
Carbon			
C(s, graphite)	0	0	5.7
C(s, diamond)	1.88	2.9	2.4
C(g)	716.7	671.3	158.1
CH ₄ (g)	−74.6	−50.5	186.3
CH ₃ Cl(g)	−81.9	−60.2	234.6
CH ₂ Cl ₂ (g)	−95.4		270.2
CH ₂ Cl ₂ (l)	−124.2	−63.2	177.8
CHCl ₃ (l)	−134.1	−73.7	201.7
CCl ₄ (g)	−95.7	−62.3	309.7
CCl ₄ (l)	−128.2	−66.4	216.4
CH ₂ O(g)	−108.6	−102.5	218.8
CH ₂ O ₂ (l, formic acid)	−425.0	−361.4	129.0
CH ₃ NH ₂ (g, methylamine)	−22.5	32.7	242.9
CH ₃ OH(l)	−238.6	−166.6	126.8
CH ₃ OH(g)	−201.0	−162.3	239.9

Substance	ΔH_f° (kJ/mol)	ΔG_f° (kJ/mol)	S° (J/mol · K)
C ₂ H ₂ (g)	227.4	209.9	200.9
C ₂ H ₄ (g)	52.4	68.4	219.3
C ₂ H ₆ (g)	−84.68	−32.0	229.2
C ₂ H ₅ OH(l)	−277.6	−174.8	160.7
C ₂ H ₅ OH(g)	−234.8	−167.9	281.6
C ₂ H ₃ Cl (g, vinyl chloride)	37.2	53.6	264.0
C ₂ H ₄ Cl ₂ (l, dichloroethane)	−166.8	−79.6	208.5
C ₂ H ₄ O (g, acetaldehyde)	−166.2	−133.0	263.8
C ₂ H ₄ O ₂ (l, acetic acid)	−484.3	−389.9	159.8
C ₃ H ₈ (g)	−103.85	−23.4	270.3
C ₃ H ₆ O (l, acetone)	−248.4	−155.6	199.8
C ₃ H ₇ OH (l, isopropanol)	−318.1		181.1
C ₄ H ₁₀ (l)	−147.3	−15.0	231.0
C ₄ H ₁₀ (g)	−125.7	−15.71	310.0
C ₆ H ₆ (l)	49.1	124.5	173.4
C ₆ H ₅ NH ₂ (l, aniline)	31.6	149.2	191.9
C ₆ H ₅ OH (s, phenol)	−165.1	−50.4	144.0
C ₆ H ₁₂ O ₆ (s, glucose)	−1273.3	−910.4	212.1
C ₁₀ H ₈ (s, naphthalene)	78.5	201.6	167.4
C ₁₂ H ₂₂ O ₁₁ (s, sucrose)	−2226.1	−1544.3	360.24
CO(g)	−110.5	−137.2	197.7
CO ₂ (g)	−393.5	−394.4	213.8
CO ₂ (aq)	−413.8	−386.0	117.6
CO ₃ ^{2−} (aq)	−677.1	−527.8	−56.9
HCO ₃ [−] (aq)	−692.0	−586.8	91.2
H ₂ CO ₃ (aq)	−699.7	−623.2	187.4
CN [−] (aq)	151	166	118
HCN(l)	108.9	125.0	112.8
HCN(g)	135.1	124.7	201.8
CS ₂ (l)	89.0	64.6	151.3
CS ₂ (g)	116.7	67.1	237.8
COCl ₂ (g)	−219.1	−204.9	283.5
C ₆₀ (s)	2327.0	2302.0	426.0
Cesium			
Cs(s)	0	0	85.2
Cs(g)	76.5	49.6	175.6
Cs ⁺ (aq)	−258.0	−292.0	132.1
CsBr(s)	−400	−387	117
CsCl(s)	−438	−414	101.2

Substance	ΔH_f° (kJ/mol)	ΔG_f° (kJ/mol)	S° (J/mol · K)
CsF(s)	−553.5	−525.5	92.8
CsI(s)	−342	−337	127
Chlorine			
Cl(g)	121.3	105.3	165.2
Cl ₂ (g)	0	0	223.1
Cl [−] (aq)	−167.1	−131.2	56.6
HCl(g)	−92.3	−95.3	186.9
HCl(aq)	−167.2	−131.2	56.5
ClO ₂ (g)	102.5	120.5	256.8
Cl ₂ O(g)	80.3	97.9	266.2
Chromium			
Cr(s)	0	0	23.8
Cr(g)	396.6	351.8	174.5
Cr ³⁺ (aq)	−1971		
CrO ₄ ^{2−} (aq)	−872.2	−717.1	44
Cr ₂ O ₃ (s)	−1139.7	−1058.1	81.2
Cr ₂ O ₇ ^{2−} (aq)	−1476	−1279	238
Cobalt			
Co(s)	0	0	30.0
Co(g)	424.7	380.3	179.5
CoO(s)	−237.9	−214.2	53.0
Co(OH) ₂ (s)	−539.7	−454.3	79.0
Copper			
Cu(s)	0	0	33.2
Cu(g)	337.4	297.7	166.4
Cu ⁺ (aq)	51.9	50.2	−26
Cu ²⁺ (aq)	64.9	65.5	−98
CuCl(s)	−137.2	−119.9	86.2
CuCl ₂ (s)	−220.1	−175.7	108.1
CuO(s)	−157.3	−129.7	42.6
CuS(s)	−53.1	−53.6	66.5
CuSO ₄ (s)	−771.4	−662.2	109.2
Cu ₂ O(s)	−168.6	−146.0	93.1
Cu ₂ S(s)	−79.5	−86.2	120.9
Fluorine			
F(g)	79.38	62.3	158.75
F ₂ (g)	0	0	202.79
F [−] (aq)	−335.35	−278.8	−13.8
HF(g)	−273.3	−275.4	173.8
Gold			
Au(s)	0	0	47.4
Au(g)	366.1	326.3	180.5
Helium			
He(g)	0	0	126.2
Hydrogen			
H(g)	218.0	203.3	114.7
H ⁺ (aq)	0	0	0

Substance	ΔH_f° (kJ/mol)	ΔG_f° (kJ/mol)	S° (J/mol · K)
H ⁺ (g)	1536.3	1517.1	108.9
H ₂ (g)	0	0	130.7
Iodine			
I(g)	106.76	70.2	180.79
I ₂ (s)	0	0	116.14
I ₂ (g)	62.42	19.3	260.69
I [−] (aq)	−56.78	−51.57	106.45
HI(g)	26.5	1.7	206.6
Iron			
Fe(s)	0	0	27.3
Fe(g)	416.3	370.7	180.5
Fe ²⁺ (aq)	−87.9	−84.94	113.4
Fe ³⁺ (aq)	−47.69	−10.54	293.3
FeCO ₃ (s)	−740.6	−666.7	92.9
FeCl ₂ (s)	−341.8	−302.3	118.0
FeCl ₃ (s)	−399.5	−334.0	142.3
FeO(s)	−272.0	−255.2	60.75
Fe(OH) ₃ (s)	−823.0	−696.5	106.7
FeS ₂ (s)	−178.2	−166.9	52.9
Fe ₂ O ₃ (s)	−824.2	−742.2	87.4
Fe ₃ O ₄ (s)	−1118.4	−1015.4	146.4
Lead			
Pb(s)	0	0	64.8
Pb(g)	195.2	162.2	175.4
Pb ²⁺ (aq)	0.92	−24.4	18.5
PbBr ₂ (s)	−278.7	−261.9	161.5
PbCO ₃ (s)	−699.1	−625.5	131.0
PbCl ₂ (s)	−359.4	−314.1	136.0
PbI ₂ (s)	−175.5	−173.6	174.9
Pb(NO ₃) ₂ (s)	−451.9		
PbO(s)	−217.3	−187.9	68.7
PbO ₂ (s)	−277.4	−217.3	68.6
PbS(s)	−100.4	−98.7	91.2
PbSO ₄ (s)	−920.0	−813.0	148.5
Lithium			
Li(s)	0	0	29.1
Li(g)	159.3	126.6	138.8
Li ⁺ (aq)	−278.47	−293.3	12.24
LiBr(s)	−351.2	−342.0	74.3
LiCl(s)	−408.6	−384.4	59.3
LiF(s)	−616.0	−587.7	35.7
LiI(s)	−270.4	−270.3	86.8
LiNO ₃ (s)	−483.1	−381.1	90.0
LiOH(s)	−487.5	−441.5	42.8
Li ₂ O(s)	−597.9	−561.2	37.6

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Substance	ΔH_f° (kJ/mol)	ΔG_f° (kJ/mol)	S° (J/mol · K)
Magnesium			
Mg(s)	0	0	32.7
Mg(g)	147.1	112.5	148.6
Mg ²⁺ (aq)	−467.0	−455.4	−137
MgCl ₂ (s)	−641.3	−591.8	89.6
MgCO ₃ (s)	−1095.8	−1012.1	65.7
MgF ₂ (s)	−1124.2	−1071.1	57.2
MgO(s)	−601.6	−569.3	27.0
Mg(OH) ₂ (s)	−924.5	−833.5	63.2
MgSO ₄ (s)	−1284.9	−1170.6	91.6
Mg ₃ N ₂ (s)	−461	−401	88
Manganese			
Mn(s)	0	0	32.0
Mn(g)	280.7	238.5	173.7
Mn ²⁺ (aq)	−219.4	−225.6	−78.8
MnO(s)	−385.2	−362.9	59.7
MnO ₂ (s)	−520.0	−465.1	53.1
MnO ₄ [−] (aq)	−529.9	−436.2	190.6
Mercury			
Hg(l)	0	0	75.9
Hg(g)	61.4	31.8	175.0
Hg ²⁺ (aq)	170.21	164.4	−36.19
Hg ₂ ²⁺ (aq)	166.87	153.5	65.74
HgCl ₂ (s)	−224.3	−178.6	146.0
HgO(s)	−90.8	−58.5	70.3
HgS(s)	−58.2	−50.6	82.4
Hg ₂ Cl ₂ (s)	−265.4	−210.7	191.6
Nickel			
Ni(s)	0	0	29.9
Ni(g)	429.7	384.5	182.2
NiCl ₂ (s)	−305.3	−259.0	97.7
NiO(s)	−239.7	−211.7	37.99
NiS(s)	−82.0	−79.5	53.0
Nitrogen			
N(g)	472.7	455.5	153.3
N ₂ (g)	0	0	191.6
NF ₃ (g)	−132.1	−90.6	260.8
NH ₃ (g)	−45.9	−16.4	192.8
NH ₃ (aq)	−80.29	−26.50	111.3
NH ₄ ⁺ (aq)	−133.26	−79.31	111.17
NH ₄ Br(s)	−270.8	−175.2	113.0
NH ₄ Cl(s)	−314.4	−202.9	94.6
NH ₄ CN(s)	0.4		
NH ₄ F(s)	−464.0	−348.7	72.0
NH ₄ HCO ₃ (s)	−849.4	−665.9	120.9
NH ₄ I(s)	−201.4	−112.5	117.0
NH ₄ NO ₃ (s)	−365.6	−183.9	151.1
NH ₄ NO ₃ (aq)	−339.9	−190.6	259.8

Substance	ΔH_f° (kJ/mol)	ΔG_f° (kJ/mol)	S° (J/mol · K)
HNO ₃ (g)	−133.9	−73.5	266.9
HNO ₃ (aq)	−207	−110.9	146
NO(g)	91.3	87.6	210.8
NO ₂ (g)	33.2	51.3	240.1
NO ₃ [−] (aq)	−206.85	−110.2	146.70
NOBr(g)	82.2	82.4	273.7
NOCl(g)	51.7	66.1	261.7
N ₂ H ₄ (l)	50.6	149.3	121.2
N ₂ H ₄ (g)	95.4	159.4	238.5
N ₂ O(g)	81.6	103.7	220.0
N ₂ O ₄ (l)	−19.5	97.5	209.2
N ₂ O ₄ (g)	9.16	99.8	304.4
N ₂ O ₅ (s)	−43.1	113.9	178.2
N ₂ O ₅ (g)	13.3	117.1	355.7
Oxygen			
O(g)	249.2	231.7	161.1
O ₂ (g)	0	0	205.2
O ₃ (g)	142.7	163.2	238.9
OH [−] (aq)	−230.02	−157.3	−10.90
H ₂ O(l)	−285.8	−237.1	70.0
H ₂ O(g)	−241.8	−228.6	188.8
H ₂ O ₂ (l)	−187.8	−120.4	109.6
H ₂ O ₂ (g)	−136.3	−105.6	232.7
Phosphorus			
P(s, white)	0	0	41.1
P(s, red)	−17.6	−12.1	22.8
P(g)	316.5	280.1	163.2
P ₂ (g)	144.0	103.5	218.1
P ₄ (g)	58.9	24.4	280.0
PCl ₃ (l)	−319.7	−272.3	217.1
PCl ₃ (g)	−287.0	−267.8	311.8
PCl ₅ (s)	−443.5		
PCl ₅ (g)	−374.9	−305.0	364.6
PF ₅ (g)	−1594.4	−1520.7	300.8
PH ₃ (g)	5.4	13.5	210.2
POCl ₃ (l)	−597.1	−520.8	222.5
POCl ₃ (g)	−558.5	−512.9	325.5
PO ₄ ^{3−} (aq)	−1277.4	−1018.7	−220.5
HPO ₄ ^{2−} (aq)	−1292.1	−1089.2	−33.5
H ₂ PO ₄ [−] (aq)	−1296.3	−1130.2	90.4
H ₃ PO ₄ (s)	−1284.4	−1124.3	110.5
H ₃ PO ₄ (aq)	−1288.3	−1142.6	158.2
P ₄ O ₆ (s)	−1640.1		
P ₄ O ₁₀ (s)	−2984	−2698	228.9
Platinum			
Pt(s)	0	0	41.6
Pt(g)	565.3	520.5	192.4

Substance	ΔH_f° (kJ/mol)	ΔG_f° (kJ/mol)	S° (J/mol · K)
Potassium			
K(s)	0	0	64.7
K(g)	89.0	60.5	160.3
K ⁺ (aq)	−252.14	−283.3	101.2
KBr(s)	−393.8	−380.7	95.9
KCN(s)	−113.0	−101.9	128.5
KCl(s)	−436.5	−408.5	82.6
KClO ₃ (s)	−397.7	−296.3	143.1
KClO ₄ (s)	−432.8	−303.1	151.0
KF(s)	−567.3	−537.8	66.6
KI(s)	−327.9	−324.9	106.3
KNO ₃ (s)	−494.6	−394.9	133.1
KOH(s)	−424.6	−379.4	81.2
KOH(aq)	−482.4	−440.5	91.6
KO ₂ (s)	−284.9	−239.4	116.7
K ₂ CO ₃ (s)	−1151.0	−1063.5	155.5
K ₂ O(s)	−361.5	−322.1	94.14
K ₂ O ₂ (s)	−494.1	−425.1	102.1
K ₂ SO ₄ (s)	−1437.8	−1321.4	175.6
Rubidium			
Rb(s)	0	0	76.8
Rb(g)	80.9	53.1	170.1
Rb ⁺ (aq)	−251.12	−283.1	121.75
RbBr(s)	−394.6	−381.8	110.0
RbCl(s)	−435.4	−407.8	95.9
RbClO ₃ (s)	−392.4	−292.0	152
RbF(s)	−557.7		
RbI(s)	−333.8	−328.9	118.4
Scandium			
Sc(s)	0	0	34.6
Sc(g)	377.8	336.0	174.8
Selenium			
Se(s, gray)	0	0	42.4
Se(g)	227.1	187.0	176.7
H ₂ Se(g)	29.7	15.9	219.0
Silicon			
Si(s)	0	0	18.8
Si(g)	450.0	405.5	168.0
SiCl ₄ (l)	−687.0	−619.8	239.7
SiF ₄ (g)	−1615.0	−1572.8	282.8
SiH ₄ (g)	34.3	56.9	204.6
SiO ₂ (s, quartz)	−910.7	−856.3	41.5
Si ₂ H ₆ (g)	80.3	127.3	272.7
Silver			
Ag(s)	0	0	42.6
Ag(g)	284.9	246.0	173.0
Ag ⁺ (aq)	105.79	77.11	73.45

Substance	ΔH_f° (kJ/mol)	ΔG_f° (kJ/mol)	S° (J/mol · K)
AgBr(s)	−100.4	−96.9	107.1
AgCl(s)	−127.0	−109.8	96.3
AgF(s)	−204.6	−185	84
AgI(s)	−61.8	−66.2	115.5
AgNO ₃ (s)	−124.4	−33.4	140.9
Ag ₂ O(s)	−31.1	−11.2	121.3
Ag ₂ S(s)	−32.6	−40.7	144.0
Ag ₂ SO ₄ (s)	−715.9	−618.4	200.4
Sodium			
Na(s)	0	0	51.3
Na(g)	107.5	77.0	153.7
Na ⁺ (aq)	−240.34	−261.9	58.45
NaBr(s)	−361.1	−349.0	86.8
NaCl(s)	−411.2	−384.1	72.1
NaCl(aq)	−407.2	−393.1	115.5
NaClO ₃ (s)	−365.8	−262.3	123.4
NaF(s)	−576.6	−546.3	51.1
NaHCO ₃ (s)	−950.8	−851.0	101.7
NaHSO ₄ (s)	−1125.5	−992.8	113.0
NaI(s)	−287.8	−286.1	98.5
NaNO ₃ (s)	−467.9	−367.0	116.5
NaNO ₃ (aq)	−447.5	−373.2	205.4
NaOH(s)	−425.8	−379.7	64.4
NaOH(aq)	−470.1	−419.2	48.2
NaO ₂ (s)	−260.2	−218.4	115.9
Na ₂ CO ₃ (s)	−1130.7	−1044.4	135.0
Na ₂ O(s)	−414.2	−375.5	75.1
Na ₂ O ₂ (s)	−510.9	−447.7	95.0
Na ₂ SO ₄ (s)	−1387.1	−1270.2	149.6
Na ₃ PO ₄ (s)	−1917	−1789	173.8
Strontium			
Sr(s)	0	0	55.0
Sr(g)	164.4	130.9	164.6
Sr ²⁺ (aq)	−545.51	−557.3	−39
SrCl ₂ (s)	−828.9	−781.1	114.9
SrCO ₃ (s)	−1220.1	−1140.1	97.1
SrO(s)	−592.0	−561.9	54.4
SrSO ₄ (s)	−1453.1	−1340.9	117.0
Sulfur			
S(s, rhombic)	0	0	32.1
S(s, monoclinic)	0.3	0.096	32.6
S(g)	277.2	236.7	167.8
S ₂ (g)	128.6	79.7	228.2
S ₈ (g)	102.3	49.7	430.9
S ^{2−} (aq)	41.8	83.7	22
SF ₆ (g)	−1220.5	−1116.5	291.5
HS [−] (aq)	−17.7	12.4	62.0

(continued on the next page)

Substance	ΔH_f° (kJ/mol)	ΔG_f° (kJ/mol)	S° (J/mol · K)
H ₂ S(g)	−20.6	−33.4	205.8
H ₂ S(aq)	−39.4	−27.7	122
SOCl ₂ (l)	−245.6		
SO ₂ (g)	−296.8	−300.1	248.2
SO ₃ (g)	−395.7	−371.1	256.8
SO ₄ ^{2−} (aq)	−909.3	−744.6	18.5
HSO ₄ [−] (aq)	−886.5	−754.4	129.5
H ₂ SO ₄ (l)	−814.0	−690.0	156.9
H ₂ SO ₄ (aq)	−909.3	−744.6	18.5
S ₂ O ₃ ^{2−} (aq)	−648.5	−522.5	67
Tin			
Sn(s, white)	0	0	51.2
Sn(s, gray)	−2.1	0.1	44.1
Sn(g)	301.2	266.2	168.5
SnCl ₄ (l)	−511.3	−440.1	258.6
SnCl ₄ (g)	−471.5	−432.2	365.8
SnO(s)	−280.7	−251.9	57.2
SnO ₂ (s)	−577.6	−515.8	49.0
Titanium			
Ti(s)	0	0	30.7
Ti(g)	473.0	428.4	180.3
TiCl ₄ (l)	−804.2	−737.2	252.3
TiCl ₄ (g)	−763.2	−726.3	353.2

Substance	ΔH_f° (kJ/mol)	ΔG_f° (kJ/mol)	S° (J/mol · K)
TiO ₂ (s)	−944.0	−888.8	50.6
Tungsten			
W(s)	0	0	32.6
W(g)	849.4	807.1	174.0
WO ₃ (s)	−842.9	−764.0	75.9
Uranium			
U(s)	0	0	50.2
U(g)	533.0	488.4	199.8
UF ₆ (s)	−2197.0	−2068.5	227.6
UF ₆ (g)	−2147.4	−2063.7	377.9
UO ₂ (s)	−1085.0	−1031.8	77.0
Vanadium			
V(s)	0	0	28.9
V(g)	514.2	754.4	182.3
Zinc			
Zn(s)	0	0	41.6
Zn(g)	130.4	94.8	161.0
Zn ²⁺ (aq)	−153.39	−147.1	−109.8
ZnCl ₂ (s)	−415.1	−369.4	111.5
ZnO(s)	−350.5	−320.5	43.7
ZnS (s, zinc blende)	−206.0	−201.3	57.7
ZnSO ₄ (s)	−982.8	−871.5	110.5

C. Aqueous Equilibrium Constants

1. Dissociation Constants for Acids at 25 °C

Name	Formula	K_{a_1}	K_{a_2}	K_{a_3}
Acetic	HC ₂ H ₃ O ₂	1.8×10^{-5}		
Acetylsalicylic	HC ₉ H ₇ O ₄	3.3×10^{-4}		
Adipic	H ₂ C ₆ H ₈ O ₄	3.9×10^{-5}	3.9×10^{-6}	
Arsenic	H ₃ AsO ₄	5.5×10^{-3}	1.7×10^{-7}	5.1×10^{-12}
Arsenous	H ₃ AsO ₃	5.1×10^{-10}		
Ascorbic	H ₂ C ₆ H ₆ O ₆	8.0×10^{-5}	1.6×10^{-12}	
Benzoic	HC ₇ H ₅ O ₂	6.5×10^{-5}		
Boric	H ₃ BO ₃	5.4×10^{-10}		
Butanoic	HC ₄ H ₇ O ₂	1.5×10^{-5}		
Carbonic	H ₂ CO ₃	4.3×10^{-7}	5.6×10^{-11}	
Chloroacetic	HC ₂ H ₂ O ₂ Cl	1.4×10^{-3}		
Chlorous	HClO ₂	1.1×10^{-2}		
Citric	H ₃ C ₆ H ₅ O ₇	7.4×10^{-4}	1.7×10^{-5}	4.0×10^{-7}
Cyanic	HCNO	2×10^{-4}		
Formic	HCHO ₂	1.8×10^{-4}		
Hydrazoic	HN ₃	2.5×10^{-5}		

Name	Formula	K_{a_1}	K_{a_2}	K_{a_3}
Hydrocyanic	HCN	4.9×10^{-10}		
Hydrofluoric	HF	3.5×10^{-4}		
Hydrogen chromate ion	HCrO ₄ [−]	3.0×10^{-7}		
Hydrogen peroxide	H ₂ O ₂	2.4×10^{-12}		
Hydrogen selenate ion	HSeO ₄ [−]	2.2×10^{-2}		
Hydrosulfuric	H ₂ S	8.9×10^{-8}	1×10^{-19}	
Hydrotelluric	H ₂ Te	2.3×10^{23}	1.6×10^{-11}	
Hypobromous	HBrO	2.8×10^{-9}		
Hypochlorous	HClO	2.9×10^{-8}		
Hypoiodous	HIO	2.3×10^{-11}		
Iodic	HIO ₃	1.7×10^{-1}		
Lactic	HC ₃ H ₅ O ₃	1.4×10^{-4}		
Maleic	H ₂ C ₄ H ₂ O ₄	1.2×10^{-2}	5.9×10^{-7}	
Malonic	H ₂ C ₃ H ₂ O ₄	1.5×10^{-3}	2.0×10^{-6}	